

Aman Pratap Singh

Mountain View, CA | aman.pratap.personal@gmail.com | +1 (240) 960-6091
linkedin.com/in/aman-pratap-singh54 | github.com/amanpratapsingh54 | amanpratapsingh54.github.io

SUMMARY

AI Engineer with 2+ years of software engineering experience building production LLM, RAG, and agentic systems. Experienced across retrieval, evaluation, model serving, and the backend infrastructure required to deploy reliable AI applications.

EDUCATION

University of Maryland

Master of Science in Applied Machine Learning

College Park, MD

August 2024 – May 2026

- **Relevant Coursework:** Machine Learning, Deep Learning, Natural Language Processing, Computer Vision.

National Institute of Technology Hamirpur

Master of Technology (M.Tech), Computer Science and Engineering

Hamirpur, India

August 2021 – May 2022

Bachelor of Technology (B.Tech), Computer Science and Engineering

August 2017 – May 2021

EXPERIENCE

University of Maryland

Teaching & Research Assistant

College Park, MD

August 2025 – May 2026

- Built **Python** evaluation tooling for an anomaly-detection research pipeline on temporal cybersecurity streams, scoring **sentence-embedding** and **clustering**-based detectors against a rules-only baseline over a **5k-event** labeled set.
- Automated correctness and model-metric checks for ML pipeline assignments with reproducible runs, mentoring **30+** graduate students.

DeepEmergence

AI Engineer

Bengaluru, India

January 2023 – July 2024

- Designed a **RAG pipeline** over 1,200+ financial PDFs, owning chunking, embedding retrieval, and re-ranking; built an **LLM-as-judge** evaluation harness against a fixed groundedness rubric, measuring a 41% reduction in unsupported responses.
- Fine-tuned **LLaMA-2 7B** on 60K financial Q&A pairs using **QLoRA**, improving top-3 answer accuracy from 43% to 59% on a 500-query held-out set; deployed the model as a **FastAPI** service behind existing authentication and rate limiting.
- Architected the asynchronous document-ingestion pipeline behind the retrieval system using **Azure Document Intelligence**, **Celery**, and MongoDB for OCR and structured extraction; scaled workers from 2 to 8, increasing throughput from 400 to 1,200 PDFs/hour.
- Owned platform performance across 8 microservices: cut p95 latency from 800 ms to 120 ms on a 5M+ row table by diagnosing N+1 queries and missing composite indexes through **PostgreSQL** query-plan analysis, and built a **Kafka-to-ClickHouse** telemetry pipeline with Datadog ingestion-lag alerting.

Pristyn Care

Software Engineer

Gurugram, India

June 2022 – December 2022

- Built **REST APIs** for appointment and clinic-management workflows and integrated a **Dialogflow** chatbot with the **WhatsApp Business API** and **Google My Business**, automating **300+ weekly** patient interactions and reducing manual support load on routine queries by **50%**.
- Trained a patient no-show forecasting model on historical scheduling data to support overbooking and reminder workflows, reducing scheduling backlog by 22% across **15+** specialties.

PROJECTS

TerpAgent: Multi-Service Campus Assistant | FastAPI, Redis, Docker, AWS

April 2026 – Present

- Designed a **FastAPI** gateway unifying 11 campus services behind **30 schema-validated endpoints**; built an **agentic orchestration layer** on the Anthropic tool-use API with structured tool schemas, multi-turn state, and **deterministic fallback routing** on tool failure.
- Implemented **async concurrent tool dispatch** for multi-step requests; deployed on **AWS EC2** with **Docker**, a **Redis** caching layer for read-heavy endpoints, and a **circuit breaker**, reducing p95 latency from **380 ms to 230 ms**.

ReviewAgent: Automated Pull Request Reviewer | Python, PostgreSQL, GitHub App, Docker, AWS

January 2026 – Present

- Built a **GitHub App** reviewing pull requests with **Claude** behind a **Postgres SKIP LOCKED** queue: the webhook verifies the signature, enqueues one row, and acknowledges in **87 ms**, clearing GitHub's 10-second timeout and preventing duplicate reviews from redeliveries.
- Tuned the review path for **precision over recall** using diff line-position mapping, **tree-sitter** scope extraction under a token budget, and hunk prioritization before any model call, keeping core logic I/O-free and covered by **309 offline tests**.

AIGateway: Multi-Provider LLM Gateway | Java, Spring Boot, Redis, PostgreSQL, Docker

December 2025

- Built a **Spring Boot** gateway unifying **4 LLM providers** behind one OpenAI-compatible endpoint with **Resilience4j** failover and **Redis** caching, tracking per-request token cost in **PostgreSQL** for per-caller spend attribution.

TECHNICAL SKILLS

Languages: Python, SQL, Java, C++

Agents & Retrieval: RAG Pipelines, FAISS (IVF-PQ), Embedding Models, Chunking & Re-ranking

ML & LLM: PyTorch, Hugging Face, LoRA/QLoRA, scikit-learn, XGBoost, LangChain, LangGraph, Claude API, OpenAI API

Evaluation & Experimentation: LLM-as-Judge, A/B Testing, Regression Suites (pytest), MLflow, Weights & Biases

Data & Infrastructure: pandas, NumPy, PostgreSQL, MongoDB, Redis, Kafka, ClickHouse, FastAPI, Spring Boot, Celery

Platform & Tooling: AWS (EC2, S3), Docker, Datadog, pytest, Git